

NileCAM81_CUOAGX

Linux Application User Manual



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Introduction to NileCAM81_CUOAGX

NileCAM81_CUOAGX is family of NileCAM cameras from e-con Systems with high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. It uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, NileCAM81_CUOAGX offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes NileCAM81_CUOAGX fast (low latency), flexible and yet reliable.

NileCAM81_CUOAGX uses Maxim's Gigabit Multimedia Serial Link (GMSL2) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable. NileCAM81_CUOAGX embeds the Maxim's serializer which serializes the high-speed image data and transmits over coaxial cable. The serializer also supports the bidirectional control data communication between the camera and host controller. The on-board electronics of NileCAM81_CUOAGX serializer recovers the power supplied from the host controller without contaminating the data transfers.

NileCAM81_CUOAGX board has been designed and developed by e-con Systems for the NVIDIA® Jetson AGX Xavier™ and NVIDIA® Jetson AGX Orin™ development kits. NileCAM81_CUOAGX is an 8 MP GMSL camera that contains a 1/1.7" AR0821 CMOS image sensor from onsemi™ along with on-board ISP. The coaxial cable is connected to serializer board through a rugged FAKRA connector. NileCAM81_CUOAGX can be directly interfaced with Jetson AGX Xavier™/Orin™ development kit through J509 connector.

The NVIDIA® Jetson AGX Xavier™/Orin™ development kit is a full-featured development platform for visual computing. It is ideal for applications requiring high computational performance in a low power envelope.

This document describes the usage of `ecam_tk1_guvcview` application and the special features of `ecam_tk1_guvcview` application when it is used with NileCAM81_CUOAGX.

Description

NileCAM81_CUOAGX can stream the resolutions and frame rates as shown in below table.

Table 1: Supported Resolutions and Frame Rates of NileCAM81_CUOAGX

Format	Resolution	Frame Rate (DAY HDR Mode)	Frame Rate (NIGHT HDR Mode)	Frame Rate (Liner Mode)
UYVY	HD	30 fps	60fps	60 fps
	FHD	30 fps	60fps	60 fps
	4K	16 fps	16fps	16 fps

Note:

- The frame rates listed in the above table can be achieved easily in manual exposure.
- In auto exposure, maximum frame rate could be achieved with maximum lighting.

The camera controls of NileCAM81_CUOAGX are as follows:

- Brightness
- Contrast
- Saturation
- White Balance (both manual and automatic modes)
- Gamma
- Gain
- Horizontal Flip
- Vertical Flip
- Sharpness
- Exposure (manual, full Field of View (FOV) auto and region of interest (ROI) based auto modes)
- ROI Window Size
- HDR
- Trigger
- Frame Sync Mode
- Trigger Mode
- Denoise
- Exposure Compensation
- Boot Mode Selection
- Powerline Frequency Selection
- Frame Rate Control
- Special Effects

The e-CAM_TK1 guvcviewer or ecam_tk1_guvcview is a simple GTK+ interface for capturing and viewing video from the devices supported on the Jetson AGX Orin™/Xavier™ development kit.

Using e-CAM_TK1 guvcviewer or ecam_tk1_guvcview application, you can perform the following:

- Enumerate and list all video devices connected.
- Display properties of video renderer.
- Change resolution and color space or compression for video stream if different resolutions are supported by the device.
- Display currently configured values of preview.
- Capture the still images and set the path where still images will be saved.
- Display the average frame rate.

All the above listed properties can be configured by attractive and easy to use Graphical User Interface (GUI).

Please refer to the *NileCAM81_CUOAGX_Release_Notes.pdf* for the compatible Linux distribution version (L4T version).

Launching the Application

This section describes how to launch the `ecam_tk1_guvcview` application.

The commands and output messages in this manual are represented by different colors as shown in below table.

Table 2: Notation of Colors

Color	Notation
Blue	Commands running in development board
Red	Output message in development board

The steps to launch the application are as follows:

1. Connect the NileCAM81_CUOAGX MIPI camera to the J509 connector on Jetson AGX Orin™/Xavier™ development kit.
2. Power ON the Jetson AGX Orin™/Xavier™ development kit.

The module drivers of NileCAM81_CUOAGX provided by e-con Systems will be loaded automatically during the board boot.

3. Run the following command to check whether the camera is initialized.

```
$ sudo dmesg | grep -i "Detected ar0821 sensor"
```

The output of above command must have an entry as shown below.

```
Detected AR0821 sensor
```

The output message indicates that the camera is initialized properly. The number of times the above entries is repeated depends on the number of cameras connected to the Jetson™ development kit.

4. Run the following command to check the presence of video node.

```
$ ls /dev/video*
```

The output message appears as shown below.

```
/dev/video*
```

Where (*) represents the index number for each camera connected and detected.

Please refer to the *ecam_tk1_guvcview_Build_and_Install_Guide.pdf* to install the application in the Linux system.

5. Run the following commands to check and set the power mode to maximum for better performance.

```
$ sudo nvpmode1 -q
$ sudo nvpmode1 -m 0
```

6. Run the following Jetson clocks command to achieve maximum frame rate before launching the ecam_tk1_guvcview application in the Jetson Orin™/Xavier AGX™ development kit.

```
$ sudo jetson_clocks
```

7. Run the following command to launch the ecam_tk1_guvcview application.

```
$ /usr/local/ecam_tk1/bin/ecam_tk1_guvcview
```

When entering the command in terminal, the screen appears as shown below.

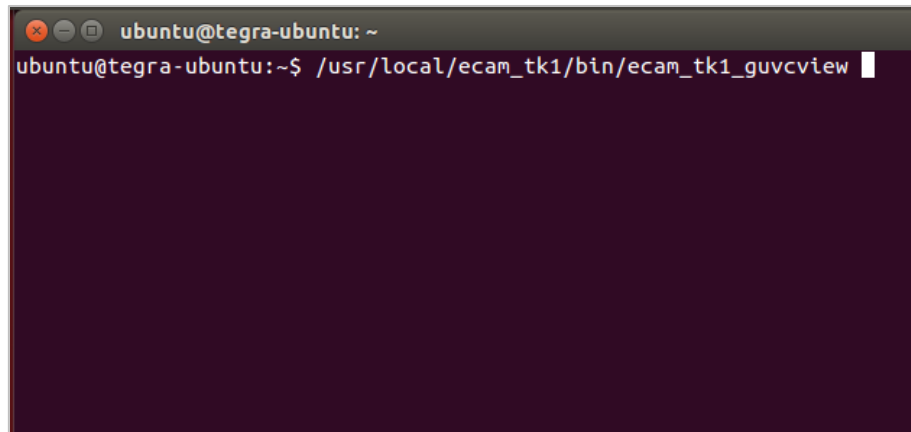


Figure 1: Terminal Window

Once the application is launched, you can view the screen similar to the screen shown below.

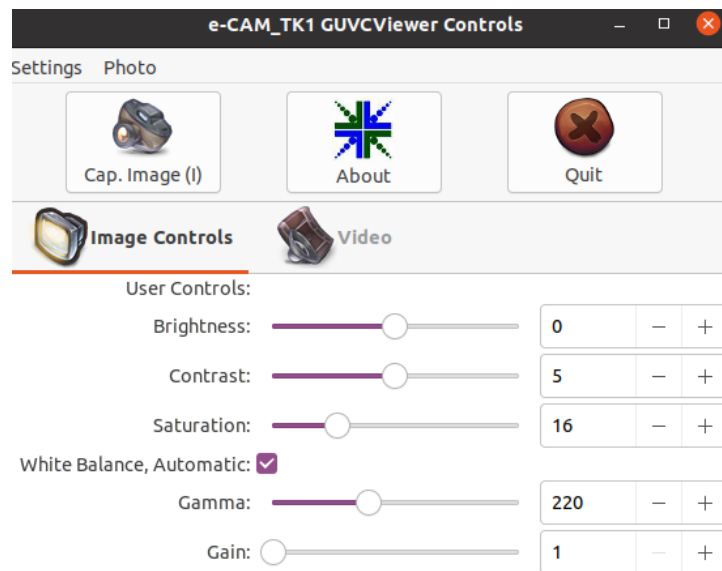


Figure 2: Initial Window Appearance

Application Features

This section describes the controls that are supported in the current version of ecam_tk1_guvview application for NileCAM81_CUOAGX.

The controls supported in ecam_tk1_guvview application are as follows:

- [Image Controls Tab](#)
- [Video Tab](#)
- [Cap. Image \(I\) Button](#)
- [About Button](#)
- [Quit Button](#)

Image Controls Tab

When selecting the **Image Controls** tab, you can view the available controls as shown below.

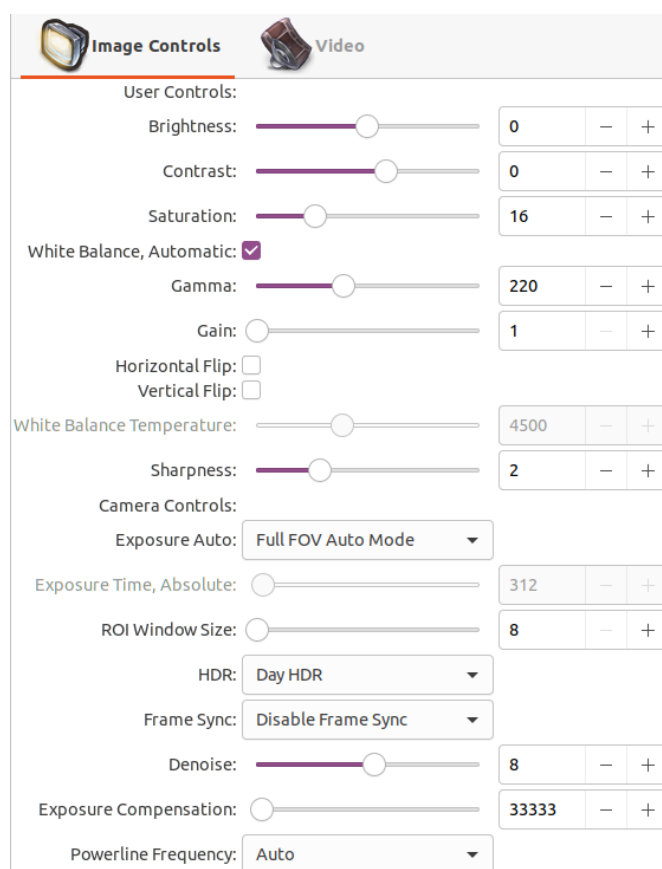


Figure 3: Image Controls Tab

The controls available in **Image Controls** tab are as follows:

- [Brightness](#)
- [Contrast](#)
- [Saturation](#)
- [White Balance](#)
- [Gamma](#)
- [Gain](#)
- [Horizontal Flip](#)
- [Vertical Flip](#)
- [Sharpness](#)
- [Exposure](#)
- [ROI Window Size](#)
- [HDR](#)
- [Trigger](#)
- [Frame Sync](#)
- [Trigger Mode](#)
- [Denoise](#)
- [Exposure Compensation](#)
- [Powerline Frequency](#)
- [Frame Rate Control](#)
- [Special Effects](#)

You can move the slider and configure the preview settings according to your needs. The sliders whose labels are not greyed can only be configured. The value being set will be displayed in the text box based on the position of the slider. As soon as the slider is moved to configure the values, the preview property will change at that instance.

The values of NileCAM81_CUOAGX controls are shown in below table.

Table 3: Values of NileCAM81_CUOAGX Controls

Controls	Minimum Value	Maximum Value	Default Value	Manual Control	Auto Control
Brightness	-15	15	0	YES	NO
Contrast	0	10	5	YES	NO
Saturation	0	60	16	YES	NO
White Balance	1000	10000	4500	YES	YES
Gamma	40	500	220	YES	NO
Gain	1	100	1	YES	NO
Horizontal Flip	0	1	0	YES	NO
Vertical Flip	0	1	0	YES	NO
Sharpness	0	7	2	YES	NO
Exposure	1 (100 μ s)	10000 (1 second)	312 (31.2 ms)	YES	YES

ROI Window Size	8	64	8	YES	NO
HDR	Day HDR	Linear	Day HDR	YES	NO
TRIGGER	0 (INTERNAL)	1 (EXTERNAL)	0 (INTERNAL)	YES	NO
Frame Sync	0 (15 HZ SYNC)	2 (60 Hz SYNC)	1 (30 Hz SYNC)	YES	NO
Trigger Mode	0 (DISABLED MODE)	1 (FREE RUNNING MODE)	2 (TRIGGER ONLY MODE)	YES	NO
Denoise	0	15	8	YES	NO
Exposure Compensation	8000	1000000	33333	YES	NO
Powerline Frequency	Auto	60HZ	Auto	YES	NO
Frame Rate Control	3	60	30	YES	NO
Special Effects	Normal	Sketch	Normal	YES	NO

Brightness

The brightness values can be changed from a minimum value of -15 to 15 by moving the slider, and the exact changes will be reflected immediately in the preview. This brightness control increases the low light performance of NileCAM81_CUOAGX. The default value is 0.

Contrast

The contrast values can be changed from a minimum value of 0 to 10 by moving the slider, and the exact changes will be reflected immediately in the preview. Increasing the contrast value increases the luminance of NileCAM81_CUOAGX. The default value is 5.

Saturation

The saturation values can be changed from a minimum value of 0 to 60 by moving the slider, and the exact changes will be reflected immediately in the preview. Increasing the saturation value increases the intensity of the colour of NileCAM81_CUOAGX. The default value is 16.

White Balance

The manual white balance can be selected by deselecting the **White Balance, Automatic** check box as shown below.

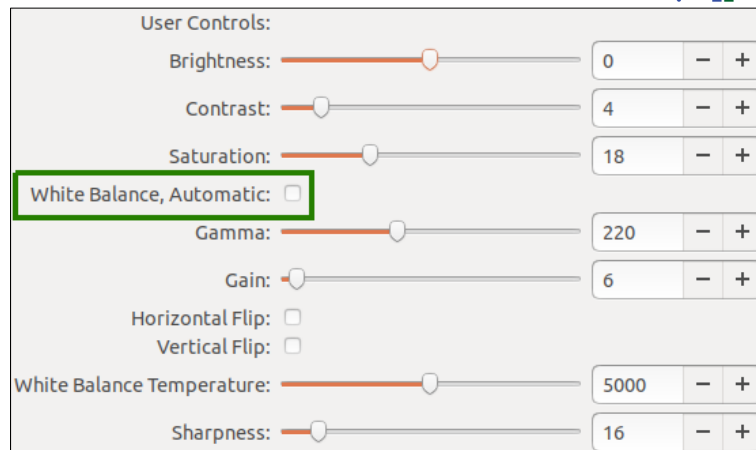


Figure 4: Manual White Balance

The manual white balance values can be changed from a minimum value of 1000 to 10000 by moving the slider of White Balance Temperature control, and the exact changes will be reflected immediately in the preview. By default White Balance Temperature is 4500.

The automatic white balance can be selected by selecting the **White Balance, Automatic** check box as shown below.

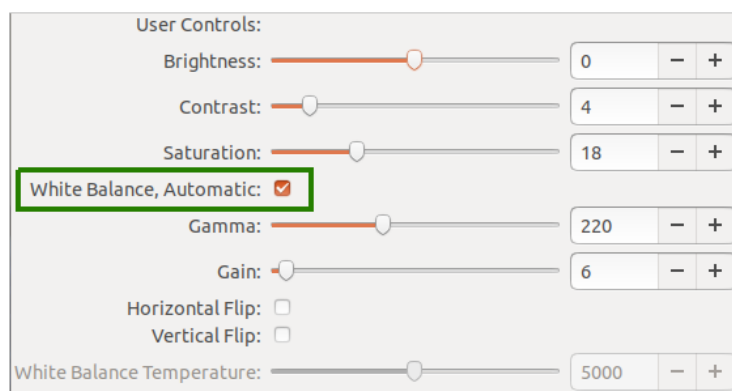


Figure 5: Automatic White Balance

Note: The White Balance Temperature control will be greyed when selecting the **White Balance, Automatic** check box.

Gamma

The gamma values can be changed from a minimum value of 40 to 500 by moving the slider, and the exact changes will be reflected immediately in the preview. The default value is 220.

Gain

The gain values can be changed from a minimum value of 1 to 100 by moving the slider. The changes are updated in the preview only when exposure control is set to manual mode. The default value is 1.

Horizontal Flip

The preview from the sensor can be horizontally flipped by selecting the **Horizontal Flip** check box. By default, this control is deselected.

Vertical Flip

The preview from the sensor can be vertically flipped by selecting the **Vertical Flip** check box. By default, this control is deselected.

Sharpness

The sharpness values can be changed from a minimum value of 0 to 7 by moving the slider, and the exact changes will be reflected immediately in the preview. This sharpness control increases the clarity of NileCAM81_CUOAGX. The default value is 2.

Exposure

NileCAM81_CUOAGX supports manual, full FOV auto and ROI based auto mode which can be controlled using the slider of Exposure Time, Absolute control of the **Image Controls** tab of ecam_tk1_guvcview. The default mode of exposure control is auto mode.

To use the slider of Exposure Time, Absolute control, you must select the **Manual Mode** from the **Exposure Auto** drop-down list box. You can view the screen similar to the screen shown below.

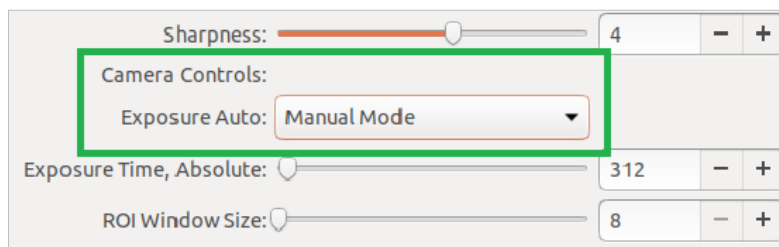


Figure 6: Manual Mode

The exposure value could be manually changed by moving the slider.

NileCAM81_CUOAGX supports exposure values ranging from 100 μ s to 1 second which is represented as 1 to 10000 in the slider. The exposure values are configured inside the CMOS image sensor based on the sensor configuration and clock configuration details. The default manual exposure value will be 312 that is 31.2ms.

To obtain a good low light performance, it is essential to change the exposure according to the change in lighting conditions. To support this feature, NileCAM81_CUOAGX has a full FOV auto mode, by which the exposure of the camera will be changed according to the lighting conditions giving the best low light performance.

When you select the **Full FOV Auto Mode** from the **Exposure Auto** drop-down list box, you can view the screen similar to the screen shown below.

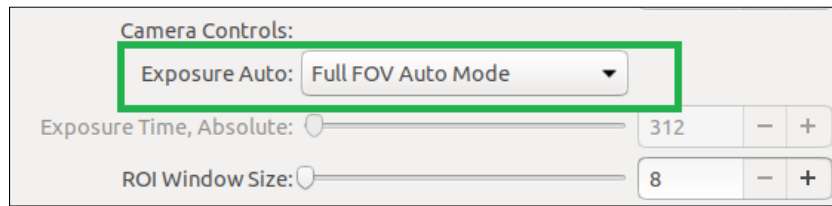


Figure 7: Full FOV Auto Mode

The exposure value applied in the sensor in ms is 1/10 of the set value as shown in below table.

Table 4: Exposure Value-Exposure Time Mapping

Exposure Value	Exposure Time
1	0.1 ms
2	0.2 ms
3	0.3 ms
..	..
10	1 ms
11	1.1 ms
12	1.2 ms
..	..
100	10 ms
..	..
1000	100 ms
..	..
7500	750 ms
10000	1 sec

Note:

- When the exposure time period is more than the time period of camera frame, the frame rate will drop.
- Controls are global across all resolutions and formats, and hence changing the control values will reflect the changes in both the formats and resolutions.
- Manual exposure is supported only in linear mode.

When you select the **ROI Based Auto Mode** from the **Exposure Auto** drop-down list box, you can view the screen similar to the screen shown below.

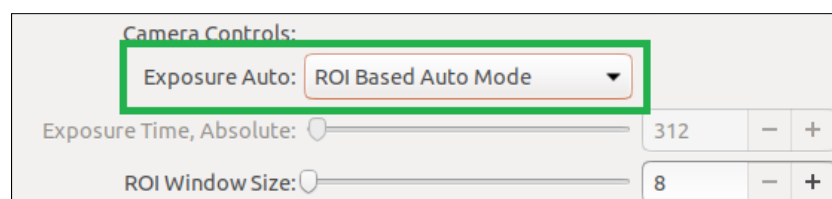


Figure 8: ROI Based Auto Mode

In ROI Based Auto Mode, you can select the ROI and at that region, the exposure value will be applied to the entire frame. You can select the required window size using the slider of ROI window size control. Click the required region on the preview window of the application and based on that region, the exposure will be applied to the entire frame.

The exposure of the image can be varied based on targeted object in the FOV. By selecting the ROI Based Auto Mode, you can also select the target object area of the frame in guvcviewer display area. The exposure algorithm will calculate the automatic exposure adjustment for the selected display area. The ROI region can be adjusted by increasing/decreasing the slider of ROI window size control.

ROI Window Size

You can change the ROI window size, by moving the slider of ROI Window size control. The default size is 8.

For window size 64, the entire frame will be the ROI. If this region exceeds or falls below the frame boundary, the ROI will be clipped automatically.

For 1280 x 720 frame size, the ROI based on the window size is shown in below table.

Table 5: Window Size - ROI Region

Window Size	ROI Region
8	1/8 (160 x 90)
16	2/8 (320 x 180)
24	3/8 (480 x 270)
32	4/8 (640 x 360)
40	5/8 (800 x 450)
48	6/8 (960 x 540)
56	7/8 (1120 x 630)
64	1 (1280 x 720)

HDR

The sensor mode selection enables to operate the camera in Day HDR/Night HDR/Linear mode. By default, module will function in Day HDR.

Day HDR: When the sensor is operated at HDR mode, the dynamic range will be greater than linear mode. It is recommended to use in a scene with good indoor light or sunlight. To enable this mode, you must select Day mode in HDR mode control.

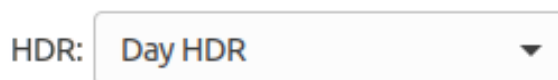


Figure 9: Day HDR Mode

- **Night HDR:** When the sensor is operated at HDR mode, the dynamic range will be greater than linear mode. It is recommended to use in a scene with low light. To enable this mode, you must select Night mode in HDR mode control.



Figure 10: Night HDR Mode

- **Linear:** When the sensor is operated at linear mode, the dynamic range of the device will be in normal range. To enable this mode, you must select Linear in HDR control.



Figure 11: Linear Mode

TRIGGER

Trigger control provide the option to switch between internal and external trigger.

- **Internal Trigger:**

By default, trigger mode is in internal trigger. To toggle trigger mode, frame sync mode should be enabled.

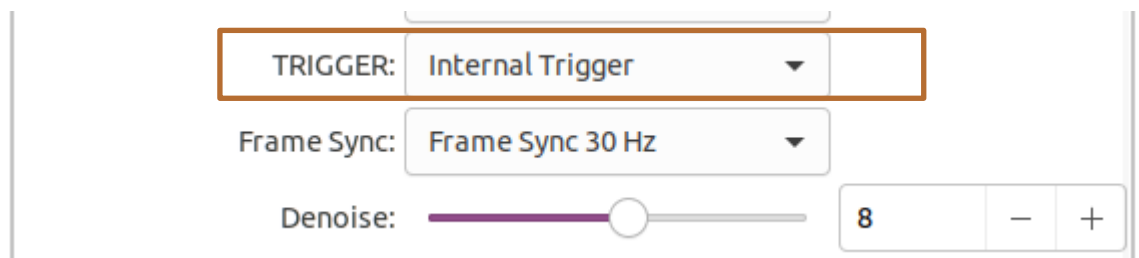


Figure 12: Internal Trigger

- **External Trigger:**

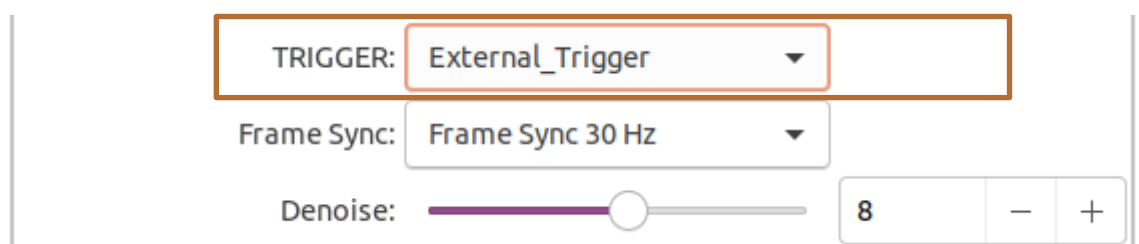


Figure 13: External Trigger

Frame Sync Mode

The frame sync control provides an option to choose Frame Sync (upto 60 Hz Sync) modes. The default mode is Frame Sync 30 Hz. When the frame sync control is enabled, the following conditions are to be satisfied:

- All camera must be in a same came mode before switching to Frame Sync mode.
- When the above conditions are not satisfied, frames may go out of sync (or) frame drops may occur.

Frame Sync 15Hz:

Cam Mode:	Day HDR
TRIGGER:	External_Trigger
Frame Sync:	Frame Sync 15 Hz

Figure 14: Frame sync 15Hz

Note: External Trigger must be enabled while using Frame Sync 15 Hz.

Frame Sync 30Hz:

Cam Mode:	Day HDR
TRIGGER:	Internal Trigger
Frame Sync:	Frame Sync 30 Hz

Figure 15: Frame sync 30Hz

Frame Sync 60Hz:

Cam Mode:	Day HDR
TRIGGER:	Internal Trigger
Frame Sync:	Frame Sync 60 Hz

Figure 16: Frame sync 60Hz

Trigger Mode

The Trigger mode selection enables to operate the camera in Disabled mode, Free running mode and Trigger only mode. By default, module will function in Free running mode.

Disabled Mode:

In Disabled mode, the camera will ignore the trigger and runs freely.

TRIGGER:	Internal Trigger
Frame Sync:	Frame Sync 15 Hz
Trigger_mode:	Disabled Mode

Figure 17: Disabled Mode

Free Running Mode:

In Free running mode, the camera frames are sync to trigger only when trigger signal is detected, otherwise runs freely.

TRIGGER:	Internal Trigger
Frame Sync:	Frame Sync 15 Hz
Trigger_mode:	Free Running Mode

Figure 18: Free Running Mode

Trigger Only Mode:

In Trigger only mode, the camera outputs the frame only when trigger signal is detected.

TRIGGER:	Internal Trigger
Frame Sync:	Frame Sync 15 Hz
Trigger_mode:	Trigger Only Mode

Figure 19: Trigger Only Mode

Denoise

The denoise control is used to remove pixel noise from the preview. This control value can be changed from a minimum value of 0 to 15 by moving the slider. The default value is 8.

Exposure Compensation

The exposure compensation control adjusts the upper limit of auto exposure. The value can be changed from a minimum value of 8000 μ s to 1000000 μ s by moving the slider. The default value is 33333 μ s.

Note: The exposure compensation control is supported in linear mode.

Powerline Frequency

The powerline frequency selection control adjusts the banding viewed in preview when exposed to artificial light sources such as tube lights or computer monitor.

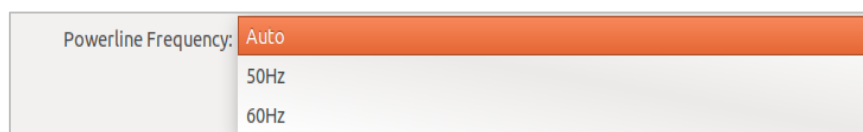


Figure 20: Powerline Frequency

Frame Rate Control

You can change the frame rate using the Frame Rate Control. The default frame rate control value will appear in the Frame Rate text box of Video Capture Pin property dialog box.

Note: The slider value of frame rate control will not have any effect above the default frame rate.

The control value will get update to its default value when the resolution/format is changed.

Note : The Frame rate control will have effect only if the frame sync option is disabled.

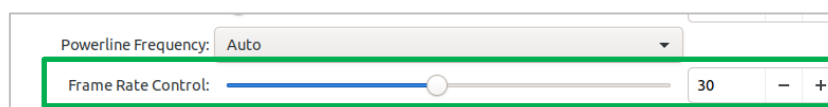


Figure 21: Frame Rate Control

Special Effects

The special effects enable four other processed image preview options in NileCAM81_CUOAGX.

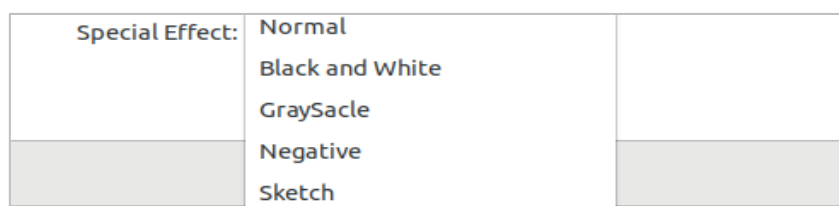


Figure 22: Special Effects

Normal: In this mode, the normal un-processed UYVY or MJPEG image stream from the camera.

Black and White: In this mode, thresholding is applied to the normal preview and the image stream is composed of black and white pixels as shown below.

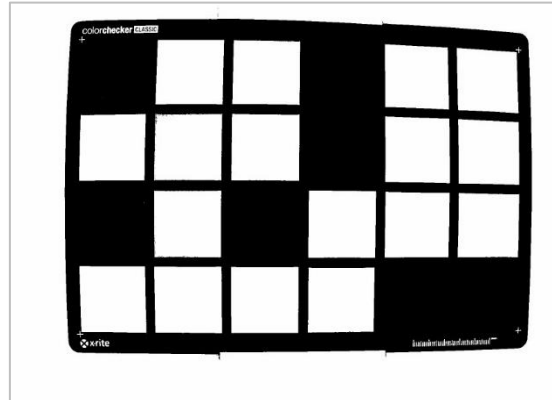


Figure 23: Black and White Effect

Grayscale: In this mode, the normal preview is desaturated, and the image stream is composed of gray shades as shown below.

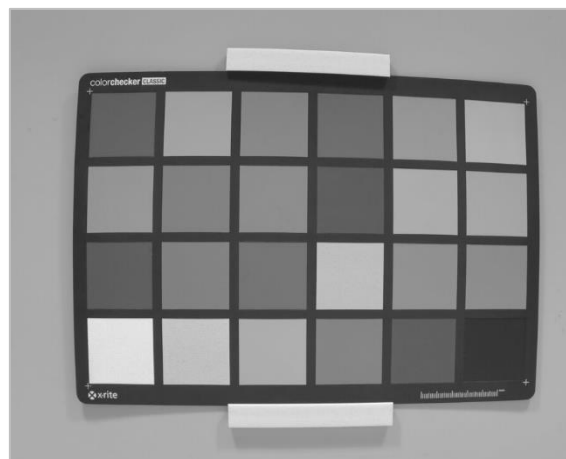


Figure 24: GrayScale Effect

Negative: In this mode, the normal preview is color inverted as shown below.



Figure 25: Negative Effect

Sketch: In this mode, an effect of edge dominant image stream useful for edge detection is produced as shown below.

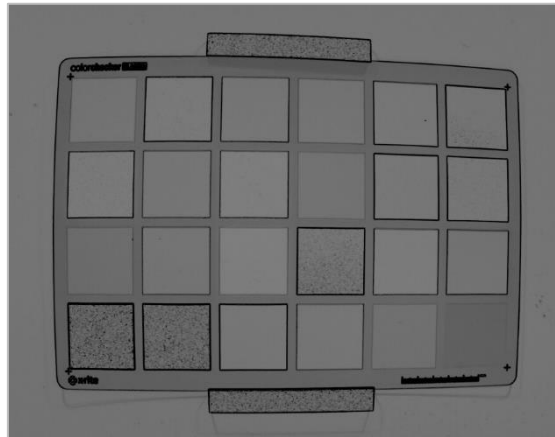


Figure 26: Sketch Effect

Video Tab

On selecting the **Video** tab, the video renderer properties will be displayed. You can adjust the video preview settings in the **Video** tab. The features supported in **Video** tab are as follows:

- [Device Selection](#)
- [Frame Rate](#)
- [Video Capture Properties](#)

Device Selection

When you click the **Device** drop-down list box, you can view the connected device as shown below.

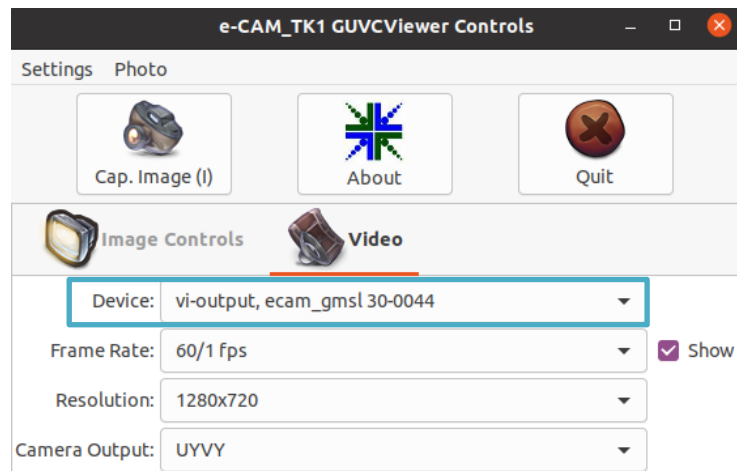


Figure 27: Device Connected

Frame Rate

The frame rate of the preview can be viewed by selecting the **Show** check box in the **Video** tab. Before selecting the **Show** check box, you can view the screen similar to the screen shown below.

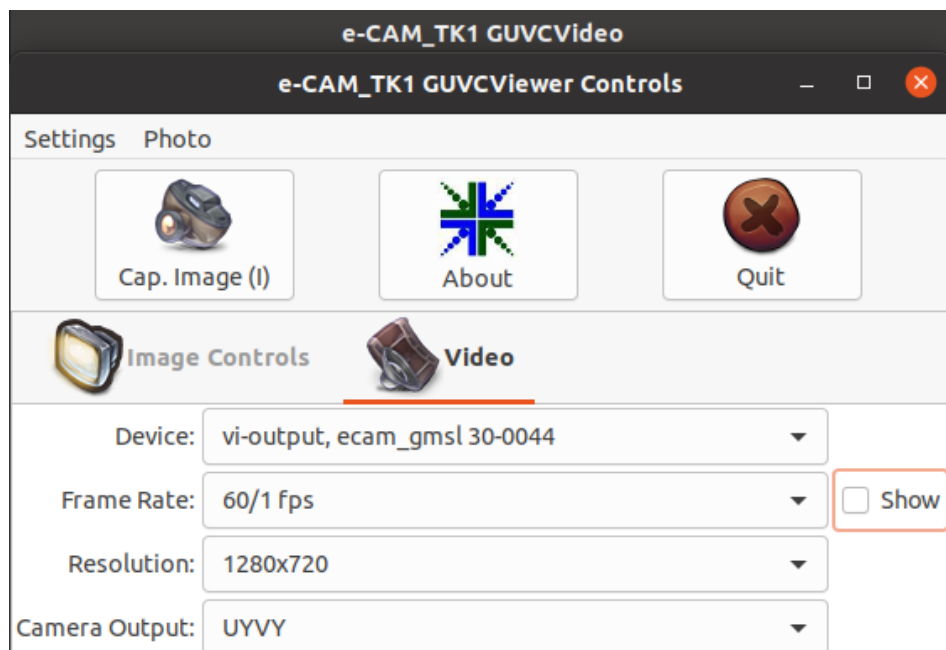


Figure 28: Selecting Show to Display Frame Rate

After selecting the **Show** check box, the frame rate will be displayed in the top of the preview window as shown below.

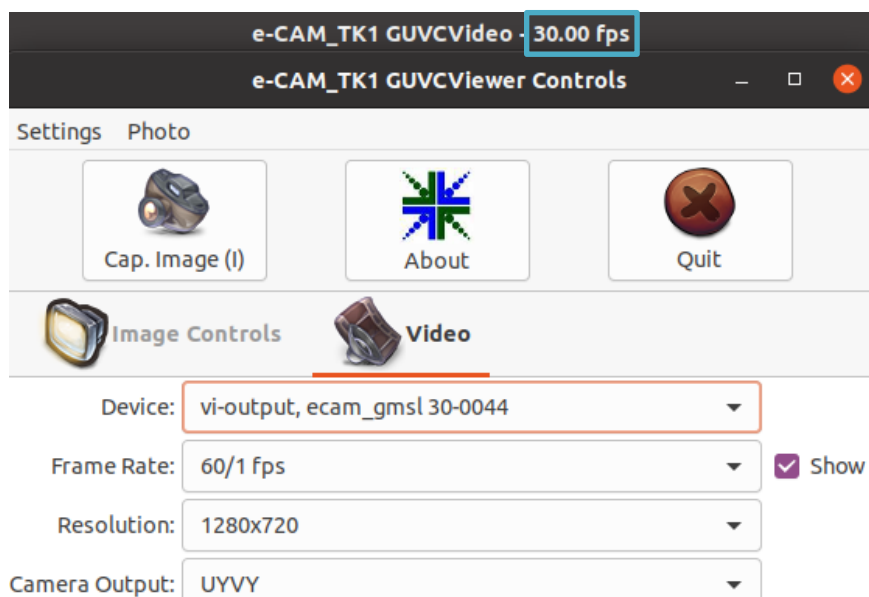


Figure 29: Displaying Frame Rate in Preview Window

Note: The frame rate will be affected by various environmental parameters.

Video Capture Properties

You can configure the output size (resolution) of stream format. Currently NileCAM81_CUOAGX supports UYVY output format. In UYVY format, three resolutions are supported. When you click the **Resolution** drop-down list box, you can view the supported resolutions as shown below.

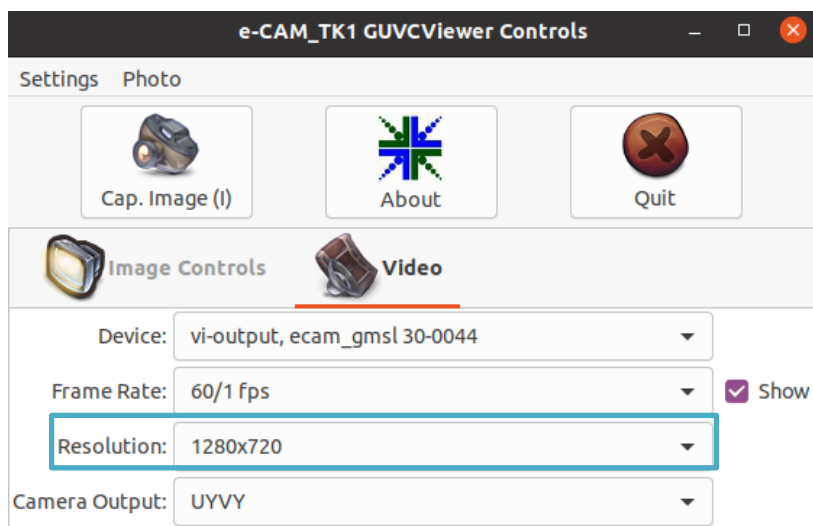


Figure 30: Supported Video Resolutions

The supported resolutions in UYVY format are listed in below table.

Table 6: Supported Format and Resolutions

Format	Resolution
UYVY	1280 x 720
	1920 x 1080
	3840 x 2160

Cap. Image (I) Button

The **Cap. Image (I)** button allows you to capture the still image of the selected device in the **Device** drop-down list box and select the path to store the image. The features supported in **Cap. Image (I)** button are as follows:

- [Still Image Capture](#)
- [Still File Path](#)

Still Image Capture

You can click the **Cap. Image (I)** button in the top of the control panel to capture the still image. You can view the screen similar to the screen shown below.

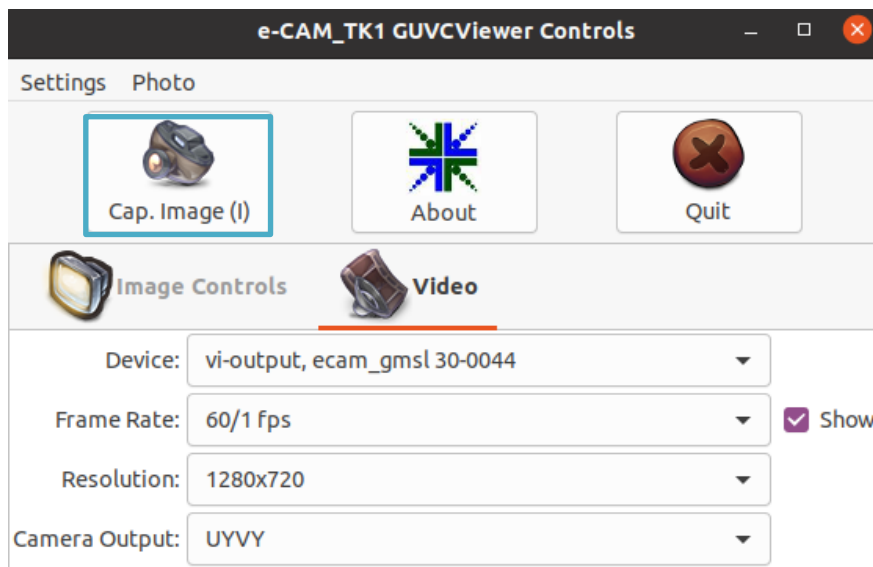


Figure 31: Capture Still Image

By default, the image will be saved in the home directory. You can configure the folder to save the images. The default name for captured image file is `guvview_image.jpg`.

Note: To capture an image of another resolution, you must switch the preview to that resolution. Unlike Windows, the Linux kernel does not support cross resolution images.

Still File Path

To create a folder for saving the captured images, click **Photo > File** as shown below.

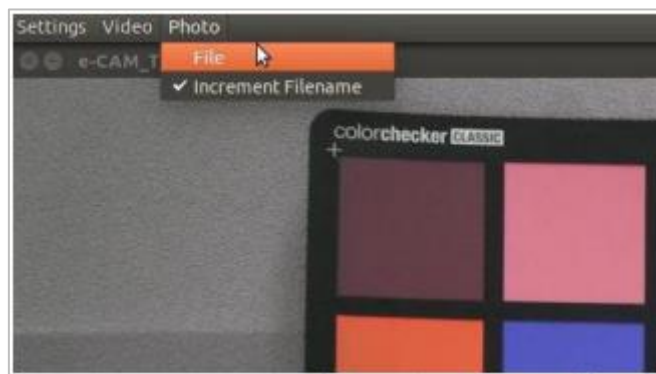


Figure 32: Save File Dialog Box to set the Still Image Folder

A **Save File** dialog box will open for selecting the file format to save the captured image. You can also browse to the required directory and set the file name, so that the captured images will also be saved in that folder.

About Button

When you click the **About** button, you can view the details of e-CAM_TK1 GUVViewer application as shown below.

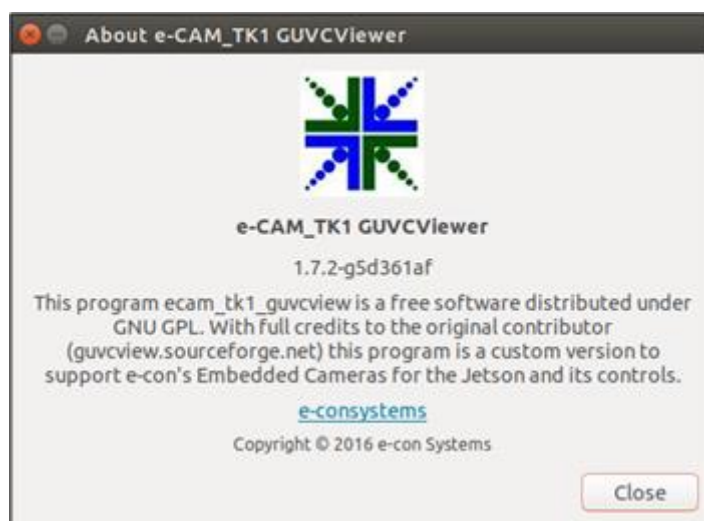


Figure 33: About e-CAM_TK1 GUVViewer

Quit Button

When you click the **Quit** button, you can exit the e-CAM_TK1 GUVViewer application.

Troubleshooting

In this section, you can view the commonly occurring issue and its troubleshooting step.

How to recover if black screen appears in `ecam_tk1_guvview` camera window?

This is a known issue. If this issue occurs, it will auto recover in few seconds and if you are still not able to recover in few second, please restart the application or change the boot mode from present mode (HDR/Linear) to next mode (Linear/HDR).

1. How can I get the updated package?

Please login to the [Developer Resources](#) website and download the latest release package.

What's Next?

After understanding the usage of `ecam_tk1_guvcview` application, you can refer to the following documents to understand more about NileCAM81_CUOAGX.

- *ecam_tk1_guvcview Build and Install Guide*
- *NileCAM81_CUOAGX Release Notes*

Glossary

API: Application Program Interface.

CMOS: Complementary Metal Oxide Semiconductor.

GMSL: Gigabit Multimedia Serial Link.

FOV: Field of View.

GIMP: GNU Image Manipulation Program.

GNU: GNU's Not Unix.

GTK: GIMP Toolkit.

GUI: Graphical User Interface.

HD: High Definition (Industry name for 1280 x 720 resolution).

L4T: Linux for Tegra.

MIPI: Mobile Industry Processor Interface.

ROI: Region of Interest.

UYVY: YUV422 16-bit image format with UYVY ordering.

VGA: Video Graphics Array (Industry name for 640 x 480 resolution).

V4L2: Video for Linux version 2 is a collection of device drivers and API for supporting real-time video capture on Linux systems.

Contact Us

If you need any support on NileCAM81_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	22-08-2022	Initial Draft for NileCAM81_CUOAGX	Camera Dev Team
1.1	23-Jan-2023	Updated sensor format from 1/2 to 1/1.7	Camera Dev Team
1.2	30-Jan-2023	Minor Changes	Camera Dev Team
1.3	10-Feb-2023	Added Sync Support	Camera Dev Team
1.4	22-Mar-2023	Updated Contrast value	Camera Dev Team
1.5	24-May-2023	Updated Frame Rate	Camera Dev Team
1.6	07-Jun-2023	Added Trigger Mode Control	Camera Dev Team
1.7	08-Jan-2024	Renamed cam mode to HDR and updated the application version	Camera Dev Team